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Cross-Dataset Comparison Report

Dataset:- Cross-Dataset (APMS) and (CSEW)

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DATA ANALYTICS INTERN

Introduction

Mental health has become a major public health concern in the United Kingdom, particularly after the COVID-19 pandemic and increasing socioeconomic stressors. Different national surveys collect information related to mental health, wellbeing, emotional distress, anxiety, treatment access, and social experiences.

This report presents a comparative analysis between two important UK datasets:

- NHS Digital Adult Psychiatric Morbidity Survey (APMS)
- Office for National Statistics Crime Survey for England and Wales (CSEW)

The objective of this analysis is to identify:

- Shared mental health patterns
- Agreements between datasets
- Conflicting findings
- Demographic similarities and differences
- Broader implications for public health understanding

Objectives of the Study

The main objectives of this cross-dataset comparison are:

1. To align APMS and CSEW datasets on shared demographic dimensions.
2. To compare mental health-related indicators across datasets.
3. To identify agreements and conflicts between survey findings.
4. To evaluate how different survey methodologies influence outcomes.
5. To generate evidence-based insights for mental health research and policy.

Dataset Overview

APMS Dataset

The Adult Psychiatric Morbidity Survey (APMS) is a nationally representative survey focusing on psychiatric disorders and mental health treatment among adults in England.

Key Features:

- Clinical mental health indicators
- Anxiety and depression prevalence
- Mental health treatment access
- Demographic variables
- Regional prevalence estimates

Strengths:

- Clinically focused
- Detailed psychiatric indicators
- Reliable mental health prevalence measures

CSEW Dataset

The Crime Survey for England and Wales (CSEW) collects information related to crime experiences, victimisation, wellbeing, and social impacts.

Key Features:

- Emotional wellbeing indicators
- Anxiety and stress measures
- Harassment and victimisation
- Social safety perceptions
- COVID-related mental health trends

Strengths:

- Large population coverage
- Social wellbeing insights
- Longitudinal trend analysis

Methodology

Data Preparation

The datasets were cleaned and standardized using Python.

Main preprocessing steps:

- Column cleaning
- Missing value handling
- Gender standardisation
- Indicator alignment
- Variable renaming
- Numeric conversion

Shared Dimensions Used

The comparison was performed using common variables available in both datasets.

Shared Dimension	APMS	CSEW
Gender	Yes	Yes
Age Group	Partial	Yes
Region	Partial	Yes
Mental Health Indicators	Yes	Partial
Anxiety/Stress Measures	Yes	Yes

Comparison Strategy

The analysis compared:

- Mental health treatment prevalence
- Emotional distress indicators
- Gender-based differences
- Relative percentage distributions

Agreement and conflict were identified using percentage difference thresholds.

Cross-Dataset Analysis

Gender-Based Comparison

The gender comparison revealed meaningful similarities between both datasets.

APMS Findings:

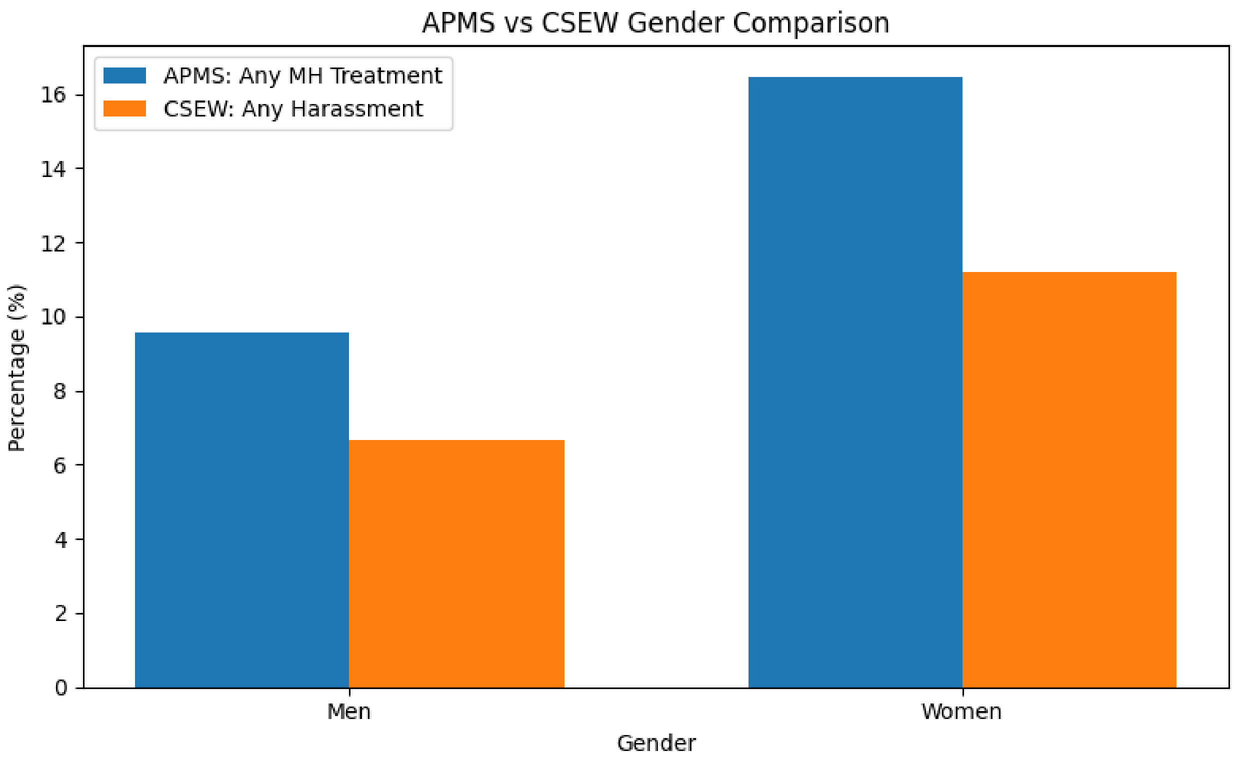
- Women showed higher mental health treatment prevalence compared to men.
- Men reported lower treatment engagement.

CSEW Findings:

- Women reported higher emotional distress and harassment-related psychological impact.
- Men showed comparatively lower prevalence.

Interpretation:

Both datasets consistently suggest that women experience greater mental health burden and psychological vulnerability.



Mental Health Treatment vs Emotional Distress

The APMS dataset focused on clinical treatment access, whereas the CSEW dataset captured social and emotional distress.

Observation:

- APMS measured formal mental health treatment usage.
- CSEW reflected social stressors and emotional outcomes.

Academic Interpretation:

Although the indicators are not identical, both datasets indicate elevated psychological burden among vulnerable groups.

Agreement Analysis:

Several patterns were consistent across both datasets.

Area	Agreement Observed
Gender differences	Women reported higher burden
Psychological vulnerability	Present in both datasets
Population inequality	Similar vulnerable groups identified
Emotional distress prevalence	Elevated in both surveys

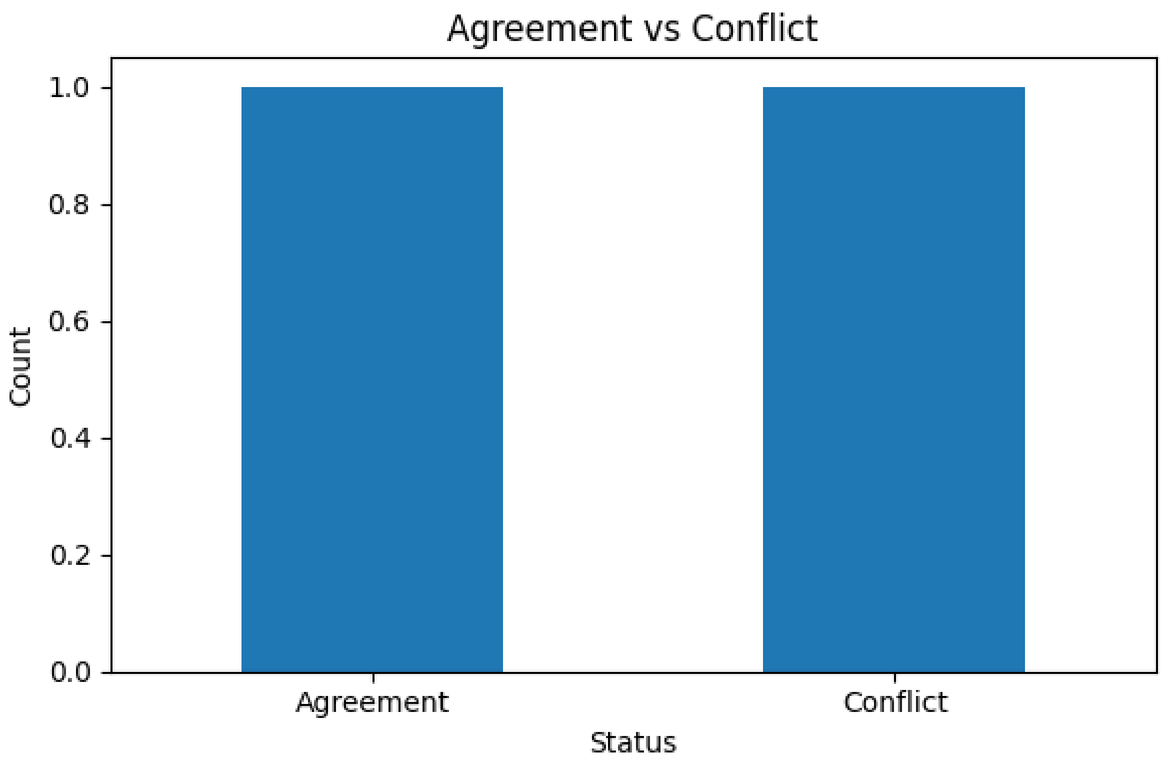
Key Insight:

Both datasets support the broader conclusion that mental health challenges are unevenly distributed across demographic groups.

5.4 Conflict Analysis

Some inconsistencies were also observed.

Area	Conflict Observed
Survey focus	Clinical vs social indicators
Measurement approach	Treatment vs emotional experience
Regional trends	Not fully aligned
Magnitude of prevalence	Different reporting scales



Possible Reasons:

1. Different survey objectives
2. Different sampling frameworks
3. Different measurement methodologies
4. Clinical vs perception-based indicators
5. Time-period variation

Discussion

The comparison demonstrates that APMS and CSEW provide complementary perspectives on mental health.

- APMS captures psychiatric morbidity and treatment access.
- CSEW captures emotional distress arising from social experiences.

Together, they provide a broader understanding of mental wellbeing in England and Wales.

The findings suggest that:

- Women remain disproportionately affected.

- Social experiences significantly influence psychological wellbeing.
- Mental health burden extends beyond clinical diagnosis.

Limitations

Several limitations should be acknowledged:

1. Different survey objectives limited direct comparability.
2. Some indicators required proxy alignment.
3. Regional variables were not perfectly matched.
4. Time periods between datasets may differ.
5. Some demographic categories required standardisation.

Conclusion

This cross-dataset comparison successfully aligned APMS and CSEW on shared dimensions to explore broader mental health patterns.

The analysis identified:

- Strong agreement regarding gender-based vulnerability
- Shared evidence of elevated emotional distress
- Important methodological differences between datasets

The study demonstrates that combining clinical and social datasets can improve understanding of mental health inequalities and support more informed public health decision-making.

Recommendations

Based on the findings, the following recommendations are suggested:

1. Improve integration of clinical and social mental health datasets.
2. Expand demographic standardisation across surveys.
3. Develop unified mental health monitoring frameworks.
4. Increase targeted support for vulnerable populations.

Tools and Technologies Used

Tool	Purpose
Python	Data cleaning and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Visualization
Excel	Data storage
Jupyter Notebook	Analysis environment

References

1. NHS Digital Adult Psychiatric Morbidity Survey (APMS)
2. Office for National Statistics Crime Survey for England and Wales (CSEW)
3. UK Mental Health Statistical Publications
4. Public Health England Mental Health Reports